STUDENT DEPRESSION ANALYSIS

A Data Analytics Project Using
Microsoft SQL Server & Tableau

Project Report

Dataset: Depression Student Dataset
Records analysed: 502
Tools: Microsoft SQL Server, Tableau, CSV

Prepared from the supplied Tableau workbook, SQL scripts, and source dataset.

1. Executive Summary

This project explores a student-related dataset to understand how recorded academic, lifestyle, and personal-context variables are distributed and how they vary across the dataset's depression label. Microsoft SQL Server is used for database setup, data standardisation, derived fields, and grouped queries. Tableau is used to present selected measures as an interactive-style dashboard of student counts.

Headline results: The source file contains **502 records** and **11 columns**. The target label is nearly evenly split: **252 records (50.2%)** are labelled "Yes" and **250 (49.8%)** are labelled "No". No blank values or exact duplicate rows were found in the supplied CSV.

The clearest descriptive patterns are higher depression-label shares at higher academic-pressure and financial-stress ratings, and lower shares at higher study-satisfaction ratings. These are associations in this dataset, not proof that any factor causes depression.

2. Project Overview and Objectives

Purpose. Build a small end-to-end analytics workflow that moves from a raw CSV dataset to structured SQL analysis and a Tableau dashboard.

Objectives

- Create and use a SQL Server database for the student dataset.
- Standardise selected categorical values and create analysis-friendly fields.
- Summarise student counts across academic, lifestyle, and stress-related variables.
- Build Tableau worksheets and combine them into a dashboard.
- Interpret the observed patterns carefully and communicate limitations.

3. Dataset Description

The supplied CSV contains 502 rows and 11 fields. The fields cover demographic information (Gender, Age), academic experience (Academic Pressure, Study Satisfaction, Study Hours), lifestyle (Sleep Duration, Dietary Habits), and other context variables (Financial Stress, Family History of Mental Illness, and a self-reported question about suicidal thoughts). The outcome field is Depression.

Dataset item	Description
Total records	502
Total columns	11
Age range	18–34 years
Average age	26.2 years
Average study hours	6.4 hours (as recorded)
Missing values in CSV	0
Exact duplicate rows	0

Gender counts: 267 male and 235 female records. Age bands used in SQL are A1 = 18–24, A2 = 25–30, and A3 = 31 years and above (the observed maximum age is 34).

4. Microsoft SQL Server: Data Preparation and Analysis

The supplied SQL scripts show a staged workflow. The database is named **Tableau Project 1**, and the imported table is referenced as **dbo.[Depression+Student+Dataset]**.

Script	Main operations evidenced in the supplied SQL
SQLQuery1.sql	Creates the database, switches context to it, and previews the imported table.
SQLQuery2.sql	Counts records by gender; changes Female/Male to F/M; checks for NULL and blank gender values.
SQLQuery3.sql	Counts records by age; adds Age_Group; assigns A1 (18–24), A2 (25–30), and A3 (other ages); counts each group.
SQLQuery4.sql	Checks table-column metadata and groups/counts records by academic pressure, study satisfaction, sleep duration, dietary
SQLQuery5.sql	Adds an identity Index_Column; converts numeric-style depression values to Yes/No; changes Depression to varchar; coun

Data preparation notes

The scripts create an index-like row identifier using an IDENTITY column, which Tableau can count as a record key. The depression field is normalised to readable Yes/No labels. The age grouping creates a compact categorical dimension for reporting. The gender update changes the stored values to F and M, while the source CSV itself uses Female and Male.

Reproducibility note: SQLQuery2 checks for NULL and blank gender values, but the supplied scripts do not show a general missing-value treatment for every column. The CSV currently has no missing values, so no imputation was required for the analysis in this report. The SQL files are scripts; the report does not claim that they were executed against a live SQL Server instance during report generation.

5. Tableau Dashboard and Visual Design

The supplied Tableau workbook contains one dashboard named **Dashboard 1** and five worksheets. The worksheet titles indicate that each view compares a selected factor with student count (SC).

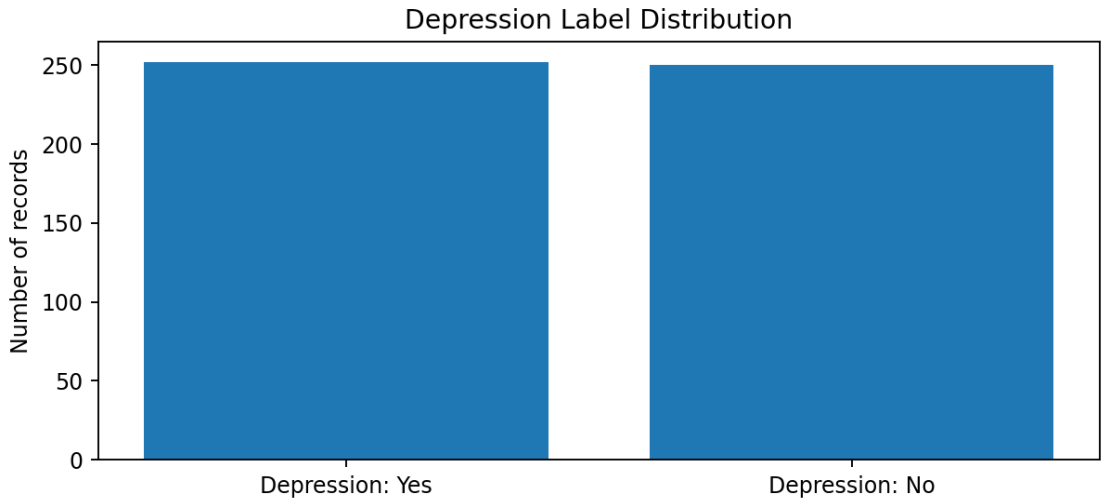
Worksheet	Measure / dimension shown
AP & SC	Academic Pressure and Student Count
FS & SC	Financial Stress and Student Count
SD & SC	Sleep Duration and Student Count
SH & SC	Study Hours and Student Count
SS & SC	Study Satisfaction and Student Count

The dashboard places the five views together: sleep duration and study hours near the top, study satisfaction across the middle, and academic pressure and financial stress along the lower section. The workbook also defines a phone layout that stacks the views vertically. The dashboard therefore gives a compact overview of how the records are distributed across key dimensions.

Interpretation boundary: The visible worksheet titles and shelves show counts by each selected factor. They do not, by themselves, demonstrate that every chart is split by the Depression outcome. The outcome-based comparisons in the next section were calculated separately from the supplied CSV for this report.

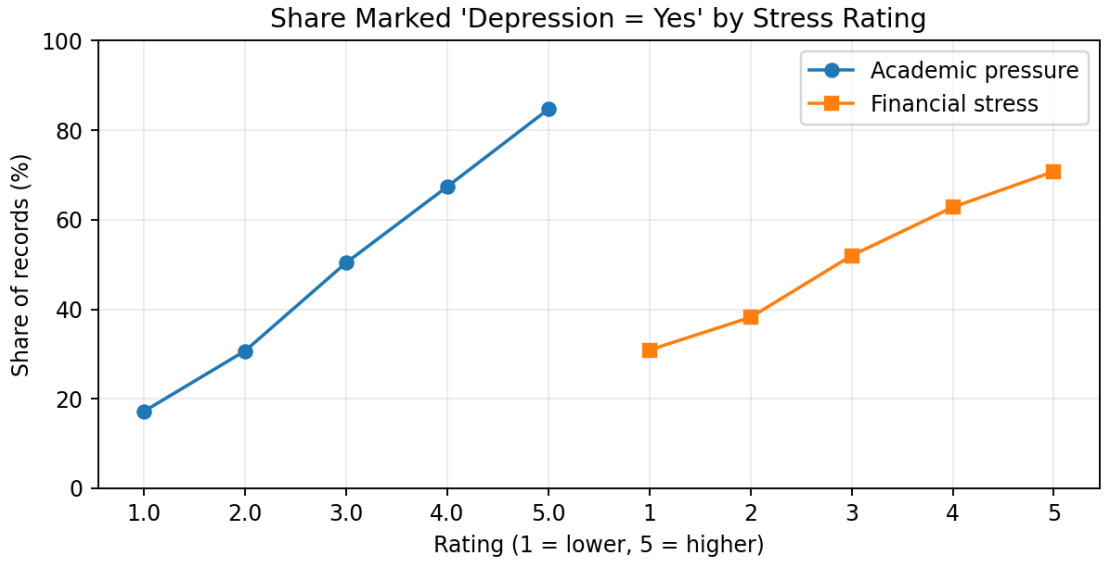
6. Exploratory Analysis and Key Findings

6.1 Overall outcome distribution



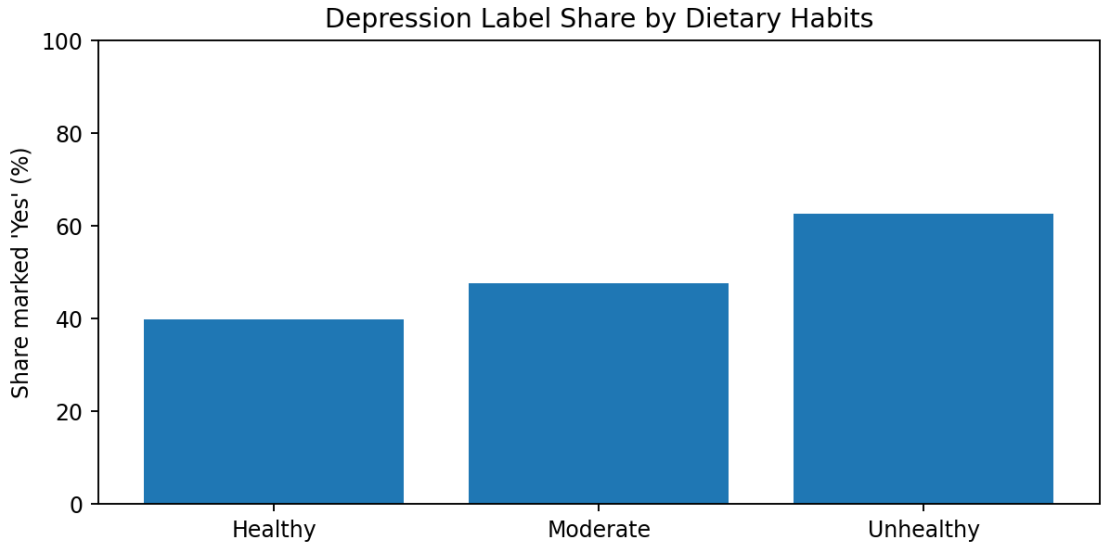
Of the 502 records, 252 (50.2%) are labelled Yes and 250 (49.8%) No. This near-even split makes it useful to compare descriptive percentages across groups, while remembering that the dataset's sampling method and representativeness are not documented.

6.2 Academic pressure and study satisfaction



The share labelled Depression = Yes rises from **17.2% at academic-pressure level 1** to **84.7% at level 5**. Conversely, it falls from **73.3% at study-satisfaction level 1** to **30.9% at level 5**. These monotonic patterns are prominent in this sample, but they should be interpreted as associations rather than causal effects.

6.3 Financial stress and dietary habits



The depression-label share increases from 30.9% at financial-stress level 1 to 70.8% at level 5. Across dietary groups, the share is 39.8% for Healthy, 47.7% for Moderate, and 62.7% for Unhealthy. Differences may reflect other variables or characteristics not captured in the dataset.

7. Additional Findings

7.1 Average values by depression label

Measure	Depression = No	Depression = Yes
Average age	27.30	25.19
Academic pressure (mean)	2.34	3.66
Study satisfaction (mean)	3.48	2.67
Study hours (mean)	5.65	7.15
Financial stress (mean)	2.50	3.35

Within the supplied records, the Yes-labelled group has higher average academic pressure, study hours, and financial stress, and lower average study satisfaction. Average age is also lower in the Yes-labelled group. These are group summaries only; they do not establish individual risk or explain why labels differ.

7.2 Other categorical comparisons

Factor group	Share labelled Yes
Suicidal-thought response = Yes	72.7%
Suicidal-thought response = No	26.0%
Family history = Yes	53.2%
Family history = No	47.5%
Sleep: less than 5 hours	52.0%
Sleep: 5–6 hours	52.0%
Sleep: 7–8 hours	52.3%
Sleep: more than 8 hours	44.5%

The response about suicidal thoughts is a sensitive self-reported variable. Its grouped comparison is included only because it is part of the supplied dataset; it must not be used to label, diagnose, or make decisions about an individual. Any real-world support or assessment should be handled by qualified professionals and appropriate safeguarding procedures.

8. Conclusion

This project demonstrates a practical analytics pipeline using Microsoft SQL Server for structured data preparation and aggregation, followed by Tableau for visual exploration. The dashboard focuses on the count distributions of academic pressure, financial stress, sleep duration, study hours, and study satisfaction. Additional analysis of the source CSV identifies descriptive relationships between the Depression label and several academic or contextual variables.

The strongest patterns in this sample are the increasing Depression = Yes share across higher academic-pressure and financial-stress ratings, and the decreasing share across higher study-satisfaction ratings. The results are useful for exploratory reporting and for demonstrating SQL-to-Tableau skills, but they should not be treated as clinical findings or causal conclusions.

9. Recommendations and Future Enhancements

• Add a Tableau view that explicitly compares Depression = Yes/No across each factor, with clear denominators and percentage labels.
• Include age-group and gender breakdowns, while avoiding small-group conclusions and protecting privacy.
• Add data-quality checks in SQL for valid category values, ranges, and unexpected nulls across all fields.
• Use descriptive labels (18–24, 25–30, 31+) instead of A1/A2/A3 in the final dashboard for easier interpretation.
• Document the source, collection method, date, and population represented by the dataset.
• If used for a research or health-related purpose, obtain appropriate ethical review and use validated measures and qualified oversight.

10. Limitations

• The dataset contains 502 records, and its collection method and representativeness are not specified in the supplied files.
• The Depression field is treated as an existing label; the project does not validate a diagnosis or assess clinical status.
• The analysis is cross-sectional/descriptive based on the provided rows; it cannot establish cause and effect.
• Potential confounding factors may not be present in the dataset.
• The Tableau workbook's five sheets display counts by selected dimensions; the workbook evidence does not show outcome-split views for every factor.
• SQL scripts were inspected as supplied. Their successful execution, server configuration, and database permissions were not independently verified.

11. Tools and Technologies

Tool	Role in the project
Microsoft SQL Server	Database creation, table inspection, value standardisation, derived age groups, identity key, and grouped S
Tableau	Five worksheets and a consolidated dashboard for student-count distributions.
CSV dataset	Source data containing 502 records and 11 variables.

12. Files Reviewed

DASHBOARD.twb — Tableau workbook; Depression+Student+Dataset.csv — source dataset; SQLQuery1.sql through SQLQuery5.sql — database setup, preparation, and analysis scripts.

End of Report
Prepared from the project files supplied in this conversation. All percentages and summary statistics in this report are calculated from the supplied CSV and rounded for presentation.